Project	Coffee Shop Sales Data Visualization Using Python, Power BI and Tableau

AIM

To analyze the Coffee Shop Sales dataset and create three visualizations—bar chart, line chart, and pie chart—using Python, Power BI, and Tableau to understand sales performance, identify popular products, and analyze sales distribution.

PROBLEM STATEMENT

Coffee shops generate large amounts of sales data through daily transactions. Understanding this data manually is difficult because it contains transaction dates, product categories, quantities, and prices. This project uses Python libraries such as Pandas and Matplotlib to process the Excel dataset and create four visualizations. These visualizations help identify high-performing product categories, observe sales trends, understand category-wise sales contributions, and analyze product price distribution.

SOFTWARE REQUIREMENTS

- Platform: Google Colab
- Programming Language: Python
- Dataset Format: Excel (.xlsx)
- Libraries: Pandas, Matplotlib, OpenPyXL
- Input Dataset: Coffee Shop Sales Excel file

DATASET DESCRIPTION

The Coffee Shop Sales dataset contains transaction details related to products sold in a coffee shop.

Column Name	Description
transaction_date	Date of the transaction
transaction_time	Time of the transaction
product_category	Category of the product sold
product_type	Specific type of product
transaction_qty	Quantity of products sold
unit_price	Price per unit

1. PYTHON

PROCEDURE

1. Import the Coffee Shop Sales dataset into Python, Power BI, and Tableau.
2. Inspect the dataset and identify the date, product category, quantity, and price columns.
3. Handle missing values and ensure that dates and numerical columns have the correct data types.
4. Calculate total sales by multiplying transaction quantity by unit price.
5. Create a bar chart to compare sales across product categories.
6. Create a line chart to analyze daily or monthly sales trends.
7. Create a pie chart to show the sales distribution across product categories.
8. Use Pandas and Matplotlib to generate the three visualizations in Python.
9. Import the dataset into Power BI and Tableau and recreate the same visualizations.
10. Compare the charts and identify sales trends, popular categories, and business insights.

```
from google.colab import files

uploaded = files.upload()
```

Choose Files Coffee_Shop_Sales.xlsx

Coffee_Shop_Sales.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 8970256 bytes, last modified: 10/18/2026 - 100% done

```
Saving Coffee_Shop_Sales.xlsx to Coffee_Shop_Sales.xlsx
```

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the uploaded Excel file
file_name = list(uploaded.keys())[0]
df = pd.read_excel(file_name)

# Display first five rows
print(df.head())

# Display column names
print(df.columns.tolist())

# Check dataset information
print(df.info())
```

```
   transaction_id transaction_date transaction_time  transaction_qty  \
0               1       2023-01-01         07:06:11                2
1               2       2023-01-01         07:08:56                2
2               3       2023-01-01         07:14:04                2
3               4       2023-01-01         07:20:24                1
4               5       2023-01-01         07:22:41                2

   store_id   store_location  product_id  unit_price    product_category  \
0         5  Lower Manhattan          32         3.0              Coffee
1         5  Lower Manhattan          57         3.1                 Tea
2         5  Lower Manhattan          59         4.5  Drinking Chocolate
3         5  Lower Manhattan          22         2.0              Coffee
4         5  Lower Manhattan          57         3.1                 Tea

            product_type               product_detail
0  Gourmet brewed coffee                  Ethiopia Rg
1        Brewed Chai tea     Spicy Eye Opener Chai Lg
2          Hot chocolate            Dark chocolate Lg
3            Drip coffee  Our Old Time Diner Blend Sm
4        Brewed Chai tea     Spicy Eye Opener Chai Lg
['transaction_id', 'transaction_date', 'transaction_time', 'transaction_qty', 'store_id', 'st
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 149116 entries, 0 to 149115
Data columns (total 11 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   transaction_id    149116 non-null  int64
 1   transaction_date  149116 non-null  datetime64[ns]
 2   transaction_time  149116 non-null  object
 3   transaction_qty   149116 non-null  int64
 4   store_id          149116 non-null  int64
 5   store_location    149116 non-null  object
 6   product_id        149116 non-null  int64
 7   unit_price        149116 non-null  float64
 8   product_category  149116 non-null  object
 9   product_type      149116 non-null  object
```

```
 10  product_detail    149116 non-null  object
dtypes: datetime64[ns](1), float64(1), int64(4), object(5)
memory usage: 12.5+ MB
None
```

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the uploaded Excel file
file_name = list(uploaded.keys())[0]
df = pd.read_excel(file_name)

# Clean column names
df.columns = df.columns.str.strip()

# Convert required columns
df["transaction_date"] = pd.to_datetime(
    df["transaction_date"], errors="coerce"
)
df["transaction_qty"] = pd.to_numeric(
    df["transaction_qty"], errors="coerce"
)
df["unit_price"] = pd.to_numeric(
    df["unit_price"], errors="coerce"
)

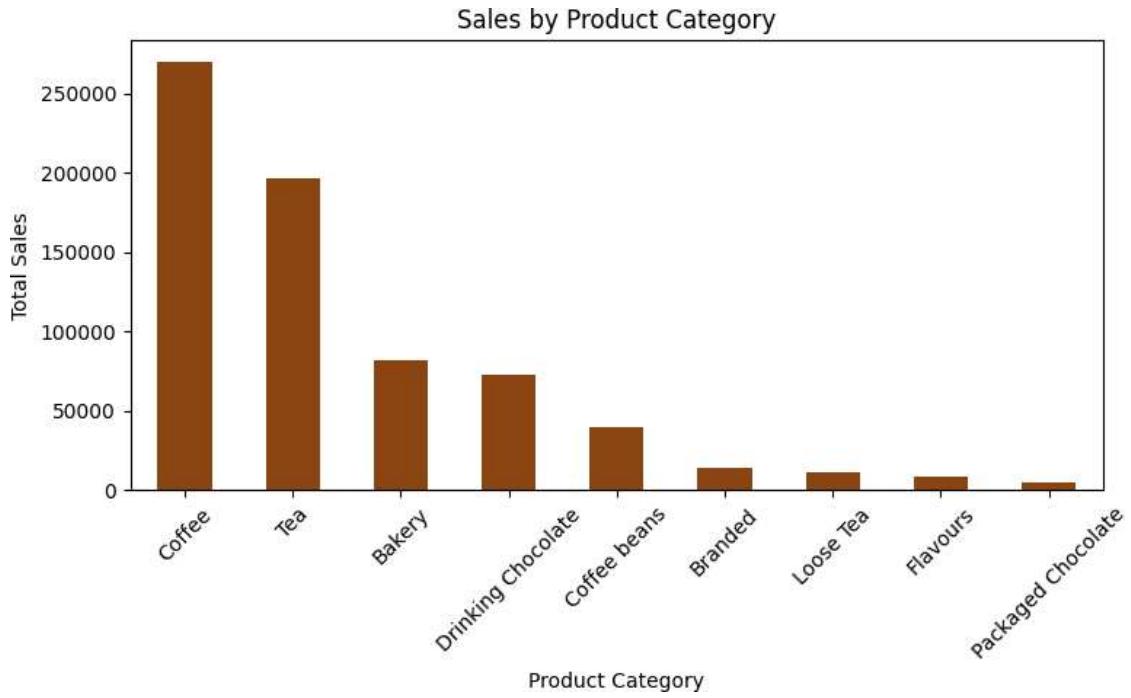
 # Remove missing values
 df = df.dropna(subset=[
     "transaction_date",
    "product_category",
    "transaction_qty",
    "unit_price"
])

# Calculate total sales
df["Total Sales"] = (
    df["transaction_qty"] * df["unit_price"]
)
```

* 1. BAR CHART - Sales by Product Category

```
# 1. BAR CHART - Sales by Product Category
category_sales = df.groupby(
    "product_category"
)["Total Sales"].sum().sort_values(ascending=False)

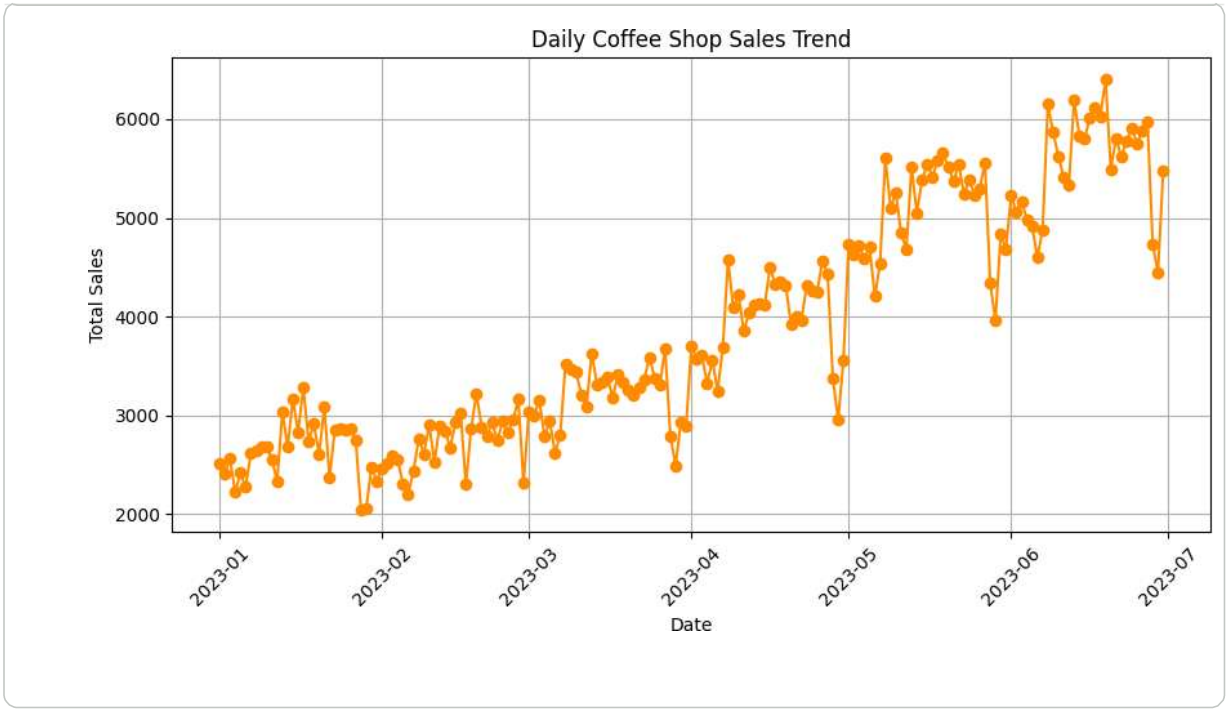
plt.figure(figsize=(8, 5))
category_sales.plot(kind="bar", color="saddlebrown")
plt.title("Sales by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



* 2. LINE CHART - Daily Sales Trend

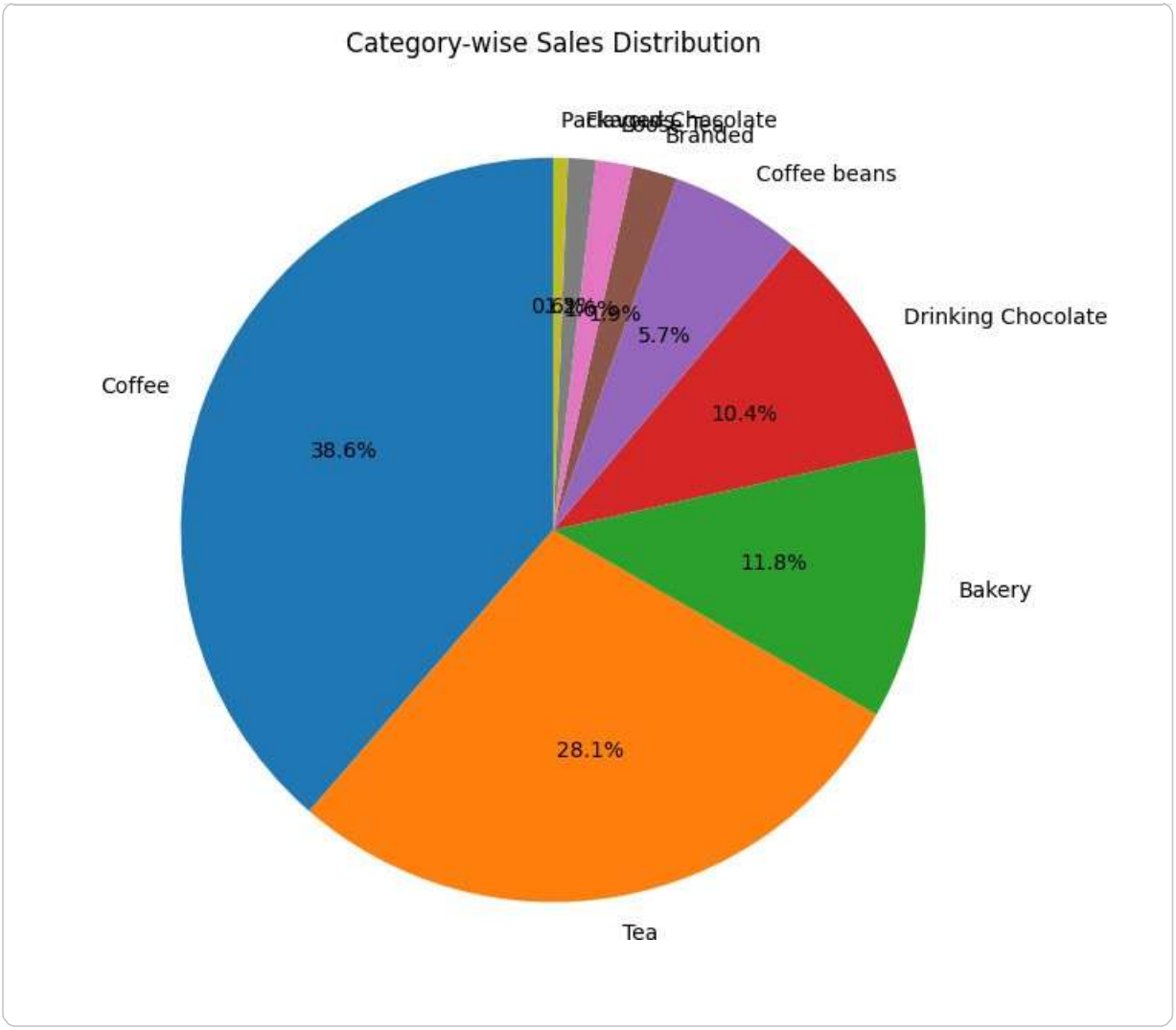
```
# 2. LINE CHART - Daily Sales Trend
daily_sales = df.groupby(
    df["transaction_date"].dt.date
)["Total Sales"].sum()

plt.figure(figsize=(9, 5))
plt.plot(daily_sales.index, daily_sales.values,
         marker="o", color="darkorange")
plt.title("Daily Coffee Shop Sales Trend")
plt.xlabel("Date")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
```



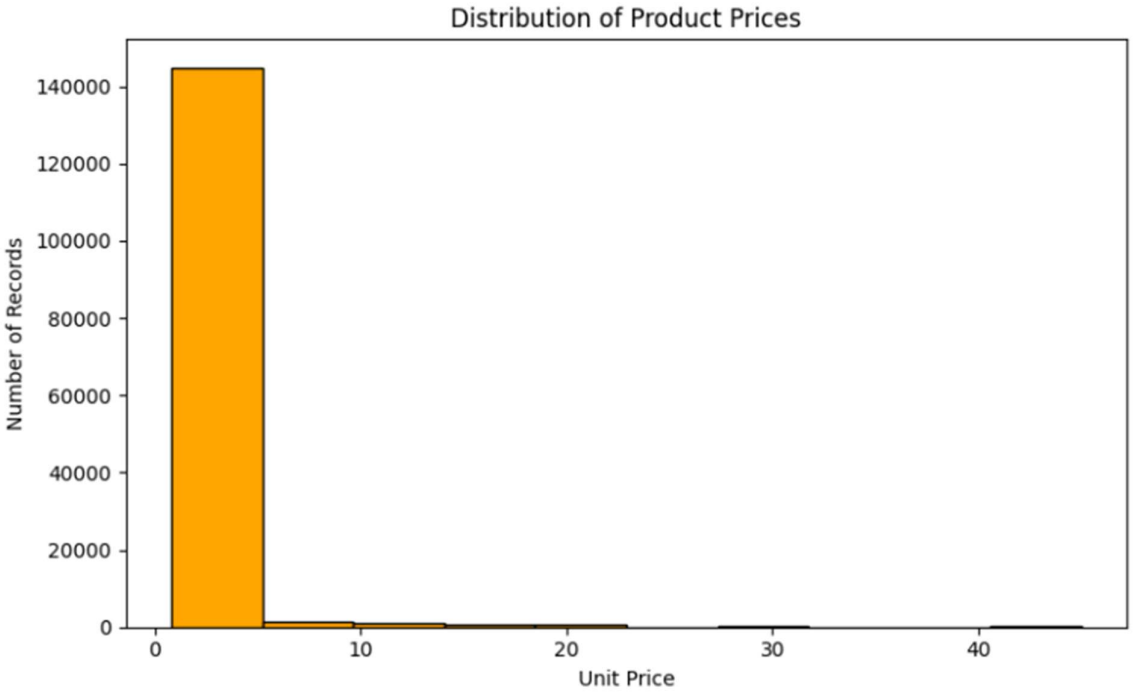
* 3. PIE CHART - Category-wise Sales Distribution

```
# 3. PIE CHART - Category-wise Sales Distribution
plt.figure(figsize=(7, 7))
plt.pie(
    category_sales.values,
    labels=category_sales.index,
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Category-wise Sales Distribution")
plt.tight_layout()
plt.show()
```



* ** 4. HISTOGRAM - Unit Price Distribution**

```
# 4. HISTOGRAM - Unit Price Distribution
plt.figure(figsize=(8, 5))
plt.hist(df["unit_price"], bins=10,
         color="orange", edgecolor="black")
plt.title("Distribution of Product Prices")
plt.xlabel("Unit Price")
plt.ylabel("Number of Records")
plt.tight_layout()
plt.show()
```


Distribution of Product Prices
Number of Records
140000
120000
100000
80000
60000
40000
20000
0
0
10
20
30
40
Unit Price

2. POWER BI

1. Open Power BI Desktop and select Home → Get Data → Text/CSV.

2. Select the Coffee Shop Sales CSV file and click Load.

3. Check the data types of transaction date, quantity, unit price, and product category.

4. Create a calculated column using Table tools → New column.

```
Total Sales = 'Coffee Shop Sales'[transaction_qty]
    * 'Coffee Shop Sales'[unit_price]
```

5. Create a bar chart by placing product_category on the X-axis and Total Sales on the Y-axis, summarized by Sum.

6. Create a line chart using transaction_date on the X-axis and Sum of Total Sales on the Y-axis.

7. Create a pie chart using product_category in Legend and Sum of Total Sales in Values.

8. Add a Card visual to display total sales.

9. Add a date or product category slicer to filter the report.

10. Arrange the visuals, format titles and colors, and save the report as Coffee_Shop_Sales.pbix.



3. TABLEAU

1. Open Tableau Desktop.

2. Under Connect, select Text File and open Student_Performance.csv.

3. Open a new worksheet.

4. Create the bar chart by placing Subject on Columns and Average Marks on Rows.

5. For the line chart, use Student Number on Columns and score values on Rows; use separate subject measures or pivot the three score columns into Subject and Score fields.

6. Create a pie chart by selecting Pie in the Marks card, placing Subject on Color and Average Marks on Angle.

7. Set the aggregation to Average where appropriate.

8. Add chart titles and customize the colors.

9. Select New Dashboard and drag all three worksheets onto the dashboard.

10. Save the workbook as Student_Performance.twbx or .twb.

VISUALIZATIONS

1. Bar Chart – Average Marks

Compares the average Mathematics, Reading, and Writing scores to identify the subject with the highest average.

2. Line Chart – Student Marks Comparison

Shows changes in marks across students and compares performance between the three subjects.

3. Pie Chart – Average Score Contribution

Displays each subject's share of the sum of the three average scores.



OBSERVATION AND INSIGHTS

- The bar chart helps identify differences between average subject scores.
- The line chart shows variation in marks among students.
- The pie chart compares each subject's proportion of the combined average scores.
- Python, Power BI, and Tableau provide different ways to create and present the same visual analysis.

RESULT

The Student Performance dataset was used to create bar charts, line charts, and pie charts using Python, Power BI, and Tableau. The visualizations support comparison of average subject marks and analysis of student performance using three different data visualization tools.